Cupcast II: A Dynamic State-Space Dixon–Coles Model with Squad-Quality Priors and Bounded Context Covariates for the 2026 FIFA World Cup

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Abstract

We replace the static Dixon–Coles model of Cupcast I with a *dynamic state-space* formulation in which each national team's latent attack and defence evolve as a Gaussian random walk over quarterly periods, so that form is a first-class quantity rather than a fixed exponential decay. The initial strength level of every team is a *learned* function of a lineup-aware squad-quality composite, built from two seasons of club performance data normalised by a club-Elo league-strength index and weighted by data-driven expected national-team minutes. A bounded, audited context layer adjusts individual match predictions for travel, host crowd, and altitude, each effect hard-capped so it cannot dominate the statistical core. Inference is by stochastic variational inference (with No-U-Turn sampling for final fits) in NumPyro [8], and match probabilities feed a 50,000-run Monte Carlo simulation of the full 48-team bracket. Held-out performance against the actual 2026 group stage, and a comparison against the Cupcast I baseline, are reported in Paper II-V.

1 Introduction

Cupcast I [4] fitted a single static rating per team by recency-weighted maximum likelihood and shrank it toward an Elo-and-squad prior. Two weaknesses motivated a rebuild. First, the squad layer was inert: an attack-only per-90 proxy with thin league coverage left player identity nearly invisible to the forecast. Second, the model was static and results-driven, which systematically overweighted recent competitive form (it priced Argentina as a 16.3% title favourite against a market consensus near 8%).

Cupcast II addresses both directly. Section 2 describes the data and the `clubform` subsystem. Section 3 specifies the dynamic state-space goal model and its clubform-informed prior. Section 4 covers inference. Section 5 specifies the bounded context layer. Section 6 describes the tournament simulation. All stochastic steps are seeded and reproduce exactly.

2 Data and the Clubform Subsystem

International results, lineups, injuries, and per-fixture player statistics are pulled from API-Football [1] (all six confederations' World Cup qualifiers, continental championships, Nations Leagues, and friendlies, 2018–2026; eighteen club leagues over two seasons), cached first and read only from cache. Youth, B, and non-FIFA selections are filtered out, and matches are held out at the tournament cutoff.

The `clubform` subsystem produces, per national team, position-aware attack/defence/goalkeeping composites. (i) A *league-strength index* normalises per-90 statistics across leagues as the median club-Elo [3] of a league's clubs, mapped to a multiplier centred on one (Premier League

1.45 down to weaker leagues near 0.65), with a curated fallback for non-European leagues club-Elo does not rate. (ii) Player quality is the league-strength-scaled, within-position z-score of attacking (non-penalty goals plus assists, shots, key passes), defensive (tackles, interceptions, blocks, duels won), and goalkeeping (save rate) output. (iii) Expected tournament minutes are estimated from *actual recent national-team lineups*: a recency-weighted start frequency reconstructs each nation's core eleven. Per-team composites join player quality to expected minutes by the global player identifier, with unmatched players imputed at the position floor; this yields ≈ 0.99 coverage for elite squads.

3 The Dynamic State-Space Goal Model

For match m at period p_m between home team h and away team a, goals are conditionally Poisson with a Dixon–Coles low-score correction. The log scoring rates are

$$\log \lambda_m = \mu + \alpha_h(p_m) - \beta_a(p_m) + \gamma \mathbb{1}[\text{host}_h], \tag{1}$$
$$\log \nu_m = \mu + \alpha_a(p_m) - \beta_h(p_m) + \gamma \mathbb{1}[\text{host}_a], \tag{2}$$

with home and away goals $x, y \sim \text{Poisson}(\lambda_m), \text{Poisson}(\nu_m)$ and the joint mass on $\{0,1\}^2$ multiplied by the correction $\tau(x, y; \lambda, \nu, \rho)$ of [4], $\rho \in (-0.3, 0.3)$. The correction is applied as an additive log-likelihood factor.

Dynamics (form). Each team's attack and defence are time-varying latent states following a non-centred Gaussian random walk across periods:

$$\alpha_i(p) = \alpha_i(0) + \sigma_\alpha \sum_{t \le p} z^\alpha_{i,t}, \qquad z^\alpha_{i,t} \sim \mathcal{N}(0, 1), \tag{3}$$

and analogously for β_i with innovation scale σ_β. The innovation scales $\sigma_\alpha, \sigma_\beta \sim \text{HalfNormal}$ are the principled replacement for Cupcast I's fixed decay half-life; they govern how fast form may move and, because the filter advances the latent state, in-tournament updating is automatic. This is the football-specific instance of the dynamic bivariate-Poisson approach of [2, 7].

Clubform-informed prior f_θ. The initial level of each team is a learned linear function of its clubform composite,

$$\alpha_i(0) \sim \mathcal{N}\big(a_0 + a_1\, \text{clubform}^\alpha_i,\ \tau^2_{\text{prior}}\big), \qquad a_0, a_1 \sim \mathcal{N}(0, \cdot), \tag{4}$$

with the defensive analogue. Data-rich teams' walks move away from this prior; data-poor teams remain anchored to it. The learned slope is positive on real data ($a_1 > 0$), confirming that club performance predicts international scoring.

4 Inference

The model is fitted in NumPyro [8] on CPU. The primary fitter is stochastic variational inference with an `AutoNormal` guide, initialised at the prior median to avoid the rate overflow a random initialisation can trigger; No-U-Turn sampling is available for final fits. Every fit threads an explicit pseudo-random key so results reproduce exactly.

5 Bounded Context Layer

Four of six context families are computed deterministically from the schedule and a static sixteen-venue table (coordinates, elevation, host nation, and how late each hosts; Mexican and Canadian venues host only through the Round of 16, with all later rounds in the United States). Per match and team we compute rest days, great-circle travel distance from the previous venue, a round-aware host flag, and venue altitude. Each maps to a hard-capped multiplicative adjustment of the scoring rate—host $\leq e^{0.12}$, travel fatigue $\geq e^{-0.10}$, short rest $\geq e^{-0.08}$, altitude on the non-adapted side $\geq e^{-0.10}$—with a per-match provenance record of every raw and applied value. The statistical core owns the probabilities; the context layer can only nudge. Morale, the one family without a structured source, is extracted in shadow mode (logged, zero weight) until an ablation proves its value.

6 Tournament Simulation

Match probabilities feed a seeded 50,000-run Monte Carlo simulation of the official 48-team structure: twelve groups of four under FIFA's group tiebreaker cascade [5], the eight best third-placed teams allocated to the Round of 32 by FIFA's Annex-C backtracking rule, and knockout rounds through the final and third-place match with extra time as rescaled rates and a goalkeeper-adjusted penalty shootout. The simulation runs in about one second and yields, for every team, the probability of each round, the title, the runner-up spot, and a podium finish. Proper scoring-rule validation [6] and the Cupcast I comparison follow in Paper II-V.

References

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